Blockchain-Integrated Frameworks for Trusted Data Sharing in Cloud of Things Networks

Casper Beck

Isobel Kent

Ellis Duffy

Andrew James

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Abstract

In the evolving landscape of the Internet of Things (IoT) and cloud computing, the convergence of these technologies has birthed the Cloud of Things (CoT) paradigm, where vast volumes of sensory data are generated, processed, and stored across distributed cloud infrastructures. Despite the remarkable scalability and flexibility of CoT networks, secure and trusted data sharing remains a paramount concern due to the distributed, heterogeneous, and resource-constrained nature of IoT devices. Traditional security mechanisms fall short in ensuring data integrity, provenance, and privacy in such decentralized environments. This paper explores the integration of blockchain technology into CoT architectures to establish robust, transparent, and tamper-resistant frameworks for secure data exchange. The research presents a comprehensive literature review of current advancements, proposes a layered architecture combining blockchain with edge-cloud processing, and evaluates the feasibility and performance of the proposed framework through experimental simulations. The findings reveal that blockchain-integrated CoT models significantly enhance trust, accountability, and data security, while posing challenges related to latency, scalability, and energy efficiency. This study contributes to the advancement of intelligent, secure, and decentralized CoT ecosystems through the strategic infusion of blockchain technologies.

Keywords: Blockchain, Cloud of Things, IoT Security, Data Sharing, Distributed Ledger Technology, Edge Computing, Trusted Frameworks, Decentralized Networks, Privacy Preservation, Smart Contracts

Introduction

The proliferation of the Internet of Things (IoT) has enabled the development of intelligent systems capable of sensing, processing, and exchanging vast volumes of data across heterogeneous devices. When combined with the power and flexibility of cloud computing, the resulting Cloud of Things (CoT) environment fosters enhanced scalability, ubiquitous access, and real-time analytics. CoT integrates the ubiquity of cloud platforms with the pervasive sensing capabilities of IoT nodes, forming a critical infrastructure for applications such as smart cities, e-health, smart grids, industrial automation, and intelligent transportation systems. However, this interconnected and decentralized data ecosystem presents significant challenges in ensuring the security, privacy, and trustworthiness of shared data.

Data exchanged across CoT networks originates from multiple untrusted sources, raising concerns regarding data integrity, provenance, access control, and confidentiality. Traditional centralized security approaches are inadequate due to the decentralized and resource-constrained nature of IoT environments. Furthermore, cloud servers may themselves become bottlenecks or points of compromise, undermining user confidence. In this context, blockchain technology emerges as a promising solution to address the trust deficit in CoT systems.

Blockchain, as a decentralized and immutable ledger, enables peer-to-peer data exchange with built-in consensus mechanisms and cryptographic assurances. It ensures that data once recorded cannot be altered retroactively, thus offering transparency and traceability. By integrating blockchain with CoT networks, it becomes possible to establish secure frameworks that provide decentralized trust, fine-grained access control, and resilient data management.

This research delves into the potential of blockchain-integrated frameworks to enable secure and trusted data sharing in CoT networks. It aims to examine existing literature, identify the gaps and limitations in current models, propose an architectural framework combining blockchain with edge-cloud paradigms, and evaluate the effectiveness of such integration in ensuring secure data exchange in distributed environments.

Literature Review

The Cloud of Things has evolved as a fusion of cloud computing and IoT, leveraging the benefits of scalability, ubiquitous access, and on-demand resource provisioning. While cloud computing offers vast storage and computational capabilities, the IoT domain contributes real-time sensory data, which often requires contextual processing at the edge of the network. The synergy of these technologies has transformed numerous application domains, but it has also introduced unprecedented security and privacy challenges. The decentralized and heterogeneous nature of CoT networks makes them particularly vulnerable to various attacks, including data tampering, spoofing, unauthorized access, and man-in-the-middle attacks.

Several researchers have addressed these concerns using conventional encryption, authentication, and access control mechanisms. However, these solutions are often unsuitable for dynamic and distributed CoT scenarios due to limitations in scalability and computational overhead. In recent years, blockchain technology has garnered significant attention for its potential to enable decentralized trust, transparency, and secure data provenance.

Notable studies have proposed the integration of blockchain in IoT systems to address issues of device authentication, secure communication, and data sharing. For instance, IBM and Samsung's ADEPT project demonstrated how blockchain could be used to establish peer-to-peer communication between IoT devices without centralized intermediaries. Similarly, other studies have explored the use of smart contracts for automating access control policies in IoT networks.

In the context of CoT, researchers have proposed various models that integrate blockchain for secure data management. One such model involves using a permissioned blockchain to manage device identities and data transactions, ensuring that only authorized entities can access and share data. Another approach involves employing lightweight consensus

mechanisms such as Practical Byzantine Fault Tolerance (PBFT) to reduce computational overhead and latency in resource-constrained environments.

Despite these advancements, challenges persist in terms of scalability, energy consumption, latency, and the integration of blockchain with edge and fog computing paradigms. Current models often overlook the dynamic nature of CoT environments and the need for real-time processing. Therefore, there is a pressing need for frameworks that combine blockchain's security features with the flexibility and responsiveness of edge-cloud architectures.

Methodology

To address the limitations identified in existing literature, this research proposes a blockchain-integrated architectural framework for trusted data sharing in CoT networks. The methodology encompasses the design, implementation, and evaluation of a layered architecture that synergizes blockchain technology with edge and cloud computing resources. The key components of the proposed framework include IoT devices, edge nodes, cloud servers, and a permissioned blockchain network.

At the lowest layer, IoT devices collect sensory data from the physical environment. Due to their limited computational and storage capabilities, these devices transmit data to nearby edge nodes for preliminary processing and validation. Edge nodes act as intermediaries that preprocess data, enforce access control policies, and interact with the blockchain network to register data hashes and access logs.

The blockchain layer operates using a permissioned model to ensure that only trusted entities participate in the consensus process. Each transaction related to data sharing or access request is recorded on the blockchain as a tamper-proof entry. Smart contracts are employed to automate access control, data provenance verification, and user authentication. This ensures that any data shared between entities is auditable, verifiable, and compliant with predefined policies.

The cloud layer provides long-term storage, advanced analytics, and large-scale processing capabilities. Data processed and validated at the edge can be securely transmitted to cloud servers for deeper analysis and storage. Access to cloud-stored data is mediated by smart contracts that enforce usage policies and record all interactions on the blockchain.

To evaluate the proposed framework, a simulation environment was developed using Hyperledger Fabric for the blockchain layer, combined with simulated IoT devices and edge nodes built using lightweight virtual machines. The framework was tested under varying workloads to assess its performance in terms of latency, throughput, energy consumption, and security effectiveness.

Results and Discussion

The experimental results demonstrate that the proposed blockchain-integrated framework significantly enhances the security and trustworthiness of data sharing in CoT networks. The use of a permissioned blockchain ensures that only authorized entities can participate in the data exchange process, thereby reducing the risk of malicious attacks and unauthorized access. The implementation of smart contracts automates access control and data verification processes, eliminating the need for centralized authorities and reducing response time.

Latency measurements revealed that the integration of edge computing with blockchain mitigates the delay typically associated with blockchain transactions. By offloading data preprocessing and validation to edge nodes, the system achieves near-real-time response for data queries and sharing operations. However, latency remains a concern when transaction volumes increase significantly, indicating the need for more efficient consensus mechanisms or layer-two scaling solutions.

Throughput analysis indicates that the framework can handle a moderate number of transactions per second without significant degradation in performance. Nevertheless, scalability remains an issue, particularly in scenarios involving large-scale IoT deployments. The use of permissioned blockchain helps alleviate some scalability issues, but further optimization is required for deployment in massive CoT environments.

Energy consumption analysis shows that while blockchain integration introduces additional computational overhead, the use of edge computing reduces the energy burden on IoT devices. This makes the architecture more sustainable and feasible for long-term deployment in energy-constrained environments.

Security analysis confirms that the blockchain-integrated framework provides robust protections against common CoT threats such as data tampering, replay attacks, and unauthorized access. The immutability and transparency of blockchain records enhance data accountability and auditability, fostering trust among stakeholders.

Despite the promising results, the study identifies several limitations. These include the complexity of smart contract design, the need for cross-chain interoperability, and the challenges of integrating legacy IoT systems with blockchain infrastructures. Future work should focus on developing lightweight blockchain protocols, dynamic smart contract management, and standardized interfaces for heterogeneous device integration.

Conclusion

The convergence of IoT and cloud computing into the Cloud of Things paradigm offers immense potential for intelligent data-driven applications. However, the distributed and dynamic nature of CoT environments introduces significant security and trust challenges, particularly in the context of data sharing across untrusted networks. This research has explored the integration of blockchain technology into CoT architectures as a means to establish secure, transparent, and decentralized frameworks for trusted data exchange.

The proposed blockchain-integrated framework leverages the strengths of edge computing and permissioned blockchain to ensure secure, scalable, and energy-efficient data sharing. Experimental evaluations validate the framework's effectiveness in enhancing data integrity, provenance, and access control, while identifying areas for further optimization in terms of latency and scalability.

As CoT networks continue to expand, the integration of blockchain technologies offers a viable path toward building resilient, intelligent, and trusted data ecosystems. The insights gained from this research can inform the development of next-generation CoT infrastructures that are secure by design and capable of supporting critical applications in healthcare, smart cities, and beyond. Further interdisciplinary collaboration and standardization efforts will be essential in realizing the full potential of blockchain-enabled CoT networks.

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